- What is the total spend per country?

Query Query History

```sql
SELECT
    m."Country",
    SUM(
        m."AmtLiq"
      + m."AmtVege"
      + m."AmtNonVeg"
      + m."AmtPes"
      + m."AmtChocolates"
      + m."AmtComm"
    ) AS total_spend
FROM marketing_data AS m
GROUP BY m."Country"
ORDER BY total_spend DESC;
```

Data Output Messages Notifications

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	Country text	total_spend numeric
1	SP	659557
2	SA	211071
3	CA	167403
4	AUS	85576
5	IND	77806
6	GER	73198
7	US	67546
8	ME	3122

- What is the total spend per product per country?

Query Query History

```sql
SELECT
    m."Country",
    SUM(m."AmtLiq")         AS total_liquor_spend,
    SUM(m."AmtVege")        AS total_veg_spend,
    SUM(m."AmtNonVeg")      AS total_nonveg_spend,
    SUM(m."AmtPes")         AS total_pes_spend,
    SUM(m."AmtChocolates")  AS total_chocolate_spend,
    SUM(m."AmtComm")        AS total_comm_spend
FROM marketing_data AS m
GROUP BY m."Country"
ORDER BY m."Country";
```

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	Country text	total_liquor_spend numeric	total_veg_spend numeric	total_nonveg_spend numeric	total_pes_spend numeric	total_chocolate_spend numeric	total_comm_spend numeric
1	AUS	42752	3689	22328	5546	4129	7132
2	CA	84066	7681	45925	9980	7607	12144
3	GER	36776	2980	20272	4601	2801	5768
4	IND	36236	3788	23729	4818	3221	6014
5	ME	1729	8	817	226	122	220
6	SA	105918	8937	58398	13670	9019	15129
7	SP	336392	28288	178409	40153	30134	46181
8	US	32214	3034	20185	4411	2863	4839

- Which products are the most popular in each country?

	Country text	most_popular_product text	total_spend numeric
1	AUS	Liquor	42752
2	CA	Liquor	84066
3	GER	Liquor	36776
4	IND	Liquor	36236
5	ME	Liquor	1729
6	SA	Liquor	105918
7	SP	Liquor	336392
8	US	Liquor	32214

```
WITH product_spend AS (
  SELECT "Country", 'Liquor' AS product, SUM("AmtLiq") AS total_spend
  FROM marketing_data
  GROUP BY "Country"

  UNION ALL
  SELECT "Country", 'Vegetables', SUM("AmtVege")
  FROM marketing_data
  GROUP BY "Country"

  UNION ALL
  SELECT "Country", 'NonVegetables', SUM("AmtNonVeg")
  FROM marketing_data
  GROUP BY "Country"
```

```
  UNION ALL
  SELECT "Country", 'Fish', SUM("AmtPes")
  FROM marketing_data
  GROUP BY "Country"

  UNION ALL
  SELECT "Country", 'Chocolates', SUM("AmtChocolates")
  FROM marketing_data
  GROUP BY "Country"

  UNION ALL
  SELECT "Country", 'Commodities', SUM("AmtComm")
  FROM marketing_data
  GROUP BY "Country"
),
ranked_products AS (
  SELECT
    "Country",
    product,
    total_spend,
    ROW_NUMBER() OVER (
      PARTITION BY "Country"
      ORDER BY total_spend DESC
    ) AS rn
  FROM product_spend
)
```

```
SELECT
   "Country",
   product AS most_popular_product,
   total_spend
FROM ranked_products
WHERE rn = 1
ORDER BY "Country";
```

Which products are most popular based on marital status

- highest total revenue per product, grouped by marital status

  - first the total revenue per product

```
-- highest total revenue per product, grouped by marital status
SELECT
   marketing_data."Marital_Status",
   SUM("AmtLiq") AS total_revenue_liquor,
   SUM("AmtVege") AS total_revenue_vege,
   SUM("AmtNonVeg") AS total_revenue_non_veg,
   SUM("AmtPes") AS total_revenue_fish,
   SUM("AmtChocolates") AS total_revenue_chocolated,
   SUM("AmtComm") AS total_revenue_comm
FROM
   marketing_data
GROUP BY
   "Marital_Status";
```

Query Query History

```sql
-- highest total revenue per product, grouped by marital status
SELECT
    marketing_data."Marital_Status",
    SUM("AmtLiq") AS total_revenue_liquor,
    SUM("AmtVege") AS total_revenue_vege,
    SUM("AmtNonVeg") AS total_revenue_non_veg,
    SUM("AmtPes") AS total_revenue_fish,
    SUM("AmtChocolates") AS total_revenue_chocolated,
    SUM("AmtComm") AS total_revenue_comm
FROM
    marketing_data
GROUP BY
    "Marital_Status";

-- Step 1: Use your query as a base to calculate total revenue for all products
WITH RevenueByStatus AS (
    SELECT
        "Marital_Status",
        SUM("AmtLiq") AS "Liquor",
        SUM("AmtVege") AS "Vegetables",
        SUM("AmtNonVeg") AS "Non-Veg",
        SUM("AmtPes") AS "Fish",
        SUM("AmtChocolates") AS "Chocolates",
```

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	Marital_Status text	total_revenue_liquor numeric	total_revenue_vege numeric	total_revenue_non_veg numeric	total_revenue_fish numeric	total_revenue_chocolated numeric	total_revenue_comm numeric
1	Widow	27902	2422	14085	3793	2878	4245
2	YOLO	644	6	100	8	6	84
3	Together	176715	14612	95374	22383	15031	24754
4	Alone	554	12	79	23	21	81
5	Absurd	711	169	625	411	61	408
6	Married	256976	21981	137888	30395	22926	36719
7	Single	137217	12840	87064	18262	12751	20397
8	Divorced	75364	6363	34848	8130	6222	10739

- and now the highest revenue product grouped by marital status

```sql
-- Step 1: Use query as a base to calculate total revenue for all products
WITH RevenueByStatus AS (
  SELECT
    "Marital_Status",
    SUM("AmtLiq") AS "Liquor",
    SUM("AmtVege") AS "Vegetables",
    SUM("AmtNonVeg") AS "Non-Veg",
    SUM("AmtPes") AS "Fish",
    SUM("AmtChocolates") AS "Chocolates",
    SUM("AmtComm") AS "Commodities"
  FROM
    marketing_data
```

```
    GROUP BY
      "Marital_Status"
  ),

  -- Step 2: Unpivot the data to compare products within each group
  UnpivotedRevenue AS (
    SELECT "Marital_Status", 'Liquor' AS product_category, "Liquor" AS revenue FROM RevenueByStatus
    UNION ALL
    SELECT "Marital_Status", 'Vegetables' AS product_category, "Vegetables" AS revenue FROM RevenueByStatus
    UNION ALL
    SELECT "Marital_Status", 'Non-Veg' AS product_category, "Non-Veg" AS revenue FROM RevenueByStatus
    UNION ALL
    SELECT "Marital_Status", 'Fish' AS product_category, "Fish" AS revenue FROM RevenueByStatus
    UNION ALL
    SELECT "Marital_Status", 'Chocolates' AS product_category, "Chocolates" AS revenue FROM RevenueByStatus
    UNION ALL
    SELECT "Marital_Status", 'Commodities' AS product_category, "Commodities" AS revenue FROM RevenueByStatus
  ),

  -- Step 3: Rank the products for each marital status by revenue
  RankedRevenue AS (
    SELECT
```

```
        "Marital_Status",
        product_category,
        revenue,
        ROW_NUMBER() OVER(PARTITION BY "Marital_Status" ORDER BY revenue DESC) as rank_num
    FROM
        UnpivotedRevenue
)

-- Step 4: Select only the top-ranked product for each status
SELECT
    "Marital_Status",
    product_category AS top_product_by_revenue,
    revenue AS highest_revenue
FROM
    RankedRevenue
WHERE
    rank_num = 1;
```

	Marital_Status text	top_product_by_revenue text	highest_revenue numeric
1	Absurd	Liquor	711
2	Alone	Liquor	554
3	Divorced	Liquor	75364
4	Married	Liquor	256976
5	Single	Liquor	137217
6	Together	Liquor	176715
7	Widow	Liquor	27902
8	YOLO	Liquor	644

- highest number of customers who made a purchase on a product (product with the most buyers), grouped by marital status

  - first, the total number of customers who made a purchase for each product grouped by marital status

Query | Query History

```sql
SELECT
    "Marital_Status",
    COUNT(DISTINCT "ID") FILTER (WHERE "AmtLiq" > 0)        AS liquor_buyers,
    COUNT(DISTINCT "ID") FILTER (WHERE "AmtVege" > 0)       AS vege_buyers,
    COUNT(DISTINCT "ID") FILTER (WHERE "AmtNonVeg" > 0)     AS nonveg_buyers,
    COUNT(DISTINCT "ID") FILTER (WHERE "AmtPes" > 0)        AS fish_buyers,
    COUNT(DISTINCT "ID") FILTER (WHERE "AmtChocolates" > 0) AS choc_buyers,
    COUNT(DISTINCT "ID") FILTER (WHERE "AmtComm" > 0)       AS comm_buyers
FROM
    marketing_data
GROUP BY
    "Marital_Status";

SELECT
    "Marital_Status",
    COUNT(*) FILTER (WHERE "AmtLiq" > 0)        AS liquor_buyers,
    COUNT(*) FILTER (WHERE "AmtVege" > 0)       AS vege_buyers,
    COUNT(*) FILTER (WHERE "AmtNonVeg" > 0)     AS nonveg_buyers,
    COUNT(*) FILTER (WHERE "AmtPes" > 0)        AS fish_buyers,
    COUNT(*) FILTER (WHERE "AmtChocolates" > 0) AS choc_buyers,
    COUNT(*) FILTER (WHERE "AmtComm" > 0)       AS comm_buyers
FROM marketing_data
GROUP BY "Marital_Status";
```

Data Output | Messages | Notifications

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	Marital_Status text	liquor_buyers bigint	vege_buyers bigint	nonveg_buyers bigint	fish_buyers bigint	choc_buyers bigint	comm_buyers bigint
1	Absurd	2	2	2	2	1	2
2	Alone	3	2	3	2	3	3
3	Divorced	229	196	231	198	193	227
4	Married	851	697	857	709	694	834
5	Single	468	402	471	390	387	460
6	Together	572	461	573	471	466	552
7	Widow	76	59	76	63	57	75
8	YOLO	2	2	2	2	2	2

- each product and which one has the most buyers

-- Step 1: Use your query as a base to calculate buyer counts for all products

```sql
WITH BuyerCounts AS (
  SELECT
    "Marital_Status",
    COUNT(*) FILTER (WHERE "AmtLiq" > 0)       AS "Liquor",
    COUNT(*) FILTER (WHERE "AmtVege" > 0)      AS "Vegetables",
    COUNT(*) FILTER (WHERE "AmtNonVeg" > 0)    AS "Non-Veg",
    COUNT(*) FILTER (WHERE "AmtPes" > 0)       AS "Fish",
    COUNT(*) FILTER (WHERE "AmtChocolates" > 0) AS "Chocolates",
    COUNT(*) FILTER (WHERE "AmtComm" > 0)      AS "Commodities"
```

```
  FROM
    marketing_data
  GROUP BY
    "Marital_Status"
),

-- Step 2: Unpivot the data to compare products within each group
UnpivotedCounts AS (
  SELECT "Marital_Status", 'Liquor' AS product_category, "Liquor" AS buyer_count FROM BuyerCounts
  UNION ALL
  SELECT "Marital_Status", 'Vegetables' AS product_category, "Vegetables" AS buyer_count FROM BuyerCounts
  UNION ALL
  SELECT "Marital_Status", 'Non-Veg' AS product_category, "Non-Veg" AS buyer_count FROM BuyerCounts
  UNION ALL
  SELECT "Marital_Status", 'Fish' AS product_category, "Fish" AS buyer_count FROM BuyerCounts
  UNION ALL
  SELECT "Marital_Status", 'Chocolates' AS product_category, "Chocolates" AS buyer_count FROM BuyerCounts
  UNION ALL
  SELECT "Marital_Status", 'Commodities' AS product_category, "Commodities" AS buyer_count FROM BuyerCounts
),

-- Step 3: Rank the products for each marital status by buyer count
```

```
RankedCounts AS (
  SELECT
    "Marital_Status",
    product_category,
    buyer_count,
    ROW_NUMBER() OVER(PARTITION BY "Marital_Status" ORDER BY buyer_count DESC)
as rank_num
  FROM
    UnpivotedCounts
)

-- Step 4: Select only the top-ranked product for each status
SELECT
  "Marital_Status",
  product_category AS top_product_by_buyers,
  buyer_count AS highest_buyer_count
FROM
  RankedCounts
WHERE
  rank_num = 1;
```

	Marital_Status text	top_product_by_buyers text	highest_buyer_count bigint
1	Absurd	Fish	2
2	Alone	Non-Veg	3
3	Divorced	Non-Veg	231
4	Married	Non-Veg	857
5	Single	Non-Veg	471
6	Together	Non-Veg	573
7	Widow	Liquor	76
8	YOLO	Liquor	2

- highest average of total revenue for a product, grouped by marital status

  - first, the average total revenue per product grouped by marital status

Query Query History

```sql
SELECT
    "Marital_Status",
    ROUND(AVG(NULLIF("AmtLiq", 0))::numeric, 2)         AS avg_spend_liquor,
    ROUND(AVG(NULLIF("AmtVege", 0))::numeric, 2)        AS avg_spend_vege,
    ROUND(AVG(NULLIF("AmtNonVeg", 0))::numeric, 2)      AS avg_spend_non_veg,
    ROUND(AVG(NULLIF("AmtPes", 0))::numeric, 2)         AS avg_spend_fish,
    ROUND(AVG(NULLIF("AmtChocolates", 0))::numeric, 2)  AS avg_spend_chocolates,
    ROUND(AVG(NULLIF("AmtComm", 0))::numeric, 2)        AS avg_spend_comm
FROM marketing_data
GROUP BY "Marital_Status"
ORDER BY "Marital_Status";
```

Data Output Messages Notifications

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	Marital_Status text	avg_spend_liquor numeric	avg_spend_vege numeric	avg_spend_non_veg numeric	avg_spend_fish numeric	avg_spend_chocolates numeric	avg_spend_comm numeric
1	Absurd	355.50	84.50	312.50	205.50	61.00	204.00
2	Alone	184.67	6.00	26.33	11.50	7.00	27.00
3	Divorced	329.10	32.46	150.86	41.06	32.24	47.31
4	Married	301.97	31.54	160.90	42.87	33.03	44.03
5	Single	293.20	31.94	184.85	46.83	32.95	44.34
6	Together	308.94	31.70	166.45	47.52	32.26	44.84
7	Widow	367.13	41.05	185.33	60.21	50.49	56.60
8	YOLO	322.00	3.00	50.00	4.00	3.00	42.00

- now the highest average spend per product grouped by marital status

```sql
WITH average_spends AS (
    SELECT
```

```
        "Marital_Status",
        AVG(NULLIF("AmtLiq", 0))        AS liquor,
        AVG(NULLIF("AmtVege", 0))       AS vegetables,
        AVG(NULLIF("AmtNonVeg", 0))     AS non_veg,
        AVG(NULLIF("AmtPes", 0))        AS fish,
        AVG(NULLIF("AmtChocolates", 0))  AS chocolates,
        AVG(NULLIF("AmtComm", 0))       AS commodities
    FROM marketing_data
    GROUP BY "Marital_Status"
  ),
  unpivoted AS (
    SELECT "Marital_Status", 'Liquor'     AS product_category, liquor     AS avg_spend FROM
average_spends
    UNION ALL
    SELECT "Marital_Status", 'Vegetables', vegetables                 FROM average_spends
    UNION ALL
    SELECT "Marital_Status", 'Non-Veg',    non_veg                    FROM average_spends
    UNION ALL
    SELECT "Marital_Status", 'Fish',      fish                       FROM average_spends
    UNION ALL
    SELECT "Marital_Status", 'Chocolates', chocolates                 FROM average_spends
    UNION ALL
    SELECT "Marital_Status", 'Commodities', commodities               FROM average_spends
  ),
  ranked AS (
    SELECT
```

```
    "Marital_Status",
    product_category,
    avg_spend,
    ROW_NUMBER() OVER (
      PARTITION BY "Marital_Status"
      ORDER BY avg_spend DESC NULLS LAST
    ) AS rn
  FROM unpivoted
)
SELECT
  "Marital_Status",
  product_category AS top_product_by_avg_spend,
  ROUND(avg_spend::numeric, 2) AS highest_avg_spend
FROM ranked
WHERE rn = 1
ORDER BY "Marital_Status";
```

	Marital_Status text	top_product_by_avg_spend text	highest_avg_spend numeric
1	Absurd	Liquor	355.50
2	Alone	Liquor	184.67
3	Divorced	Liquor	329.10
4	Married	Liquor	301.97
5	Single	Liquor	293.20
6	Together	Liquor	308.94
7	Widow	Liquor	367.13
8	YOLO	Liquor	322.00

Which products are the most popular based on whether or not there are children or teens in the home?

```
WITH household_groups AS (
  SELECT
    CASE
      WHEN "Kidhome" = 0 AND "Teenhome" = 0
        THEN 'No children or teens'
      ELSE 'Children or teens present'
    END AS household_type,
    "AmtLiq", "AmtVege", "AmtNonVeg", "AmtPes", "AmtChocolates", "AmtComm"
  FROM marketing_data
),
averages AS (
  SELECT
    household_type,
    AVG(NULLIF("AmtLiq", 0))       AS liquor,
    AVG(NULLIF("AmtVege", 0))      AS vegetables,
    AVG(NULLIF("AmtNonVeg", 0))    AS non_veg,
    AVG(NULLIF("AmtPes", 0))       AS fish,
    AVG(NULLIF("AmtChocolates", 0)) AS chocolates,
    AVG(NULLIF("AmtComm", 0))      AS commodities
  FROM household_groups
  GROUP BY household_type
),
unpivoted AS (
  SELECT household_type, 'Liquor' AS product, liquor AS avg_spend FROM averages
```

```
  UNION ALL SELECT household_type, 'Vegetables', vegetables FROM averages
  UNION ALL SELECT household_type, 'Non-Veg', non_veg FROM averages
  UNION ALL SELECT household_type, 'Fish', fish FROM averages
  UNION ALL SELECT household_type, 'Chocolates', chocolates FROM averages
  UNION ALL SELECT household_type, 'Commodities', commodities FROM averages
),
ranked AS (
  SELECT
    household_type,
    product,
    avg_spend,
    ROW_NUMBER() OVER (
      PARTITION BY household_type
      ORDER BY avg_spend DESC NULLS LAST
    ) AS rn
  FROM unpivoted
)
SELECT
  household_type,
  product AS most_popular_product,
  ROUND(avg_spend, 2) AS avg_spend
FROM ranked
WHERE rn = 1;
```

	household_type text	most_popular_product text	avg_spend numeric
1	Children or teens present	Liquor	232.66
2	No children or teens	Liquor	494.32